Guerfi Ahmed Yacine.

Cyber Security Engineering Student

AVAILABLE FOR INTERNSHIP · 2026–27

ay.guerfi@esi-sba.dz
n4khla.github.io
github.com/N4khlA
linkedin.com/in/guerfi-ahmed-yacine-6831b32b8

Final-year cyber security engineering student at ESI Sidi Bel Abbès. I learn security by building the vulnerable system, exploiting it, instrumenting it and then proving the fix. I have shipped three open-source tools covering web exploitation, ML-based intrusion detection and mobile static analysis. I'm looking for an internship in penetration testing, application security or SOC / detection engineering.

EDUCATION

École Supérieure en Informatique (ESI-SBA) · Sidi Bel Abbès, Algeria — Expected 2027

State Engineering Degree in Computer Science, Cyber Security specialisation, and Master's Degree

- Final year: offensive and defensive security coursework, hands-on ethical hacking labs, end-of-studies project

SECURITY PROJECTS

Race Condition Exploiter · Offensive / Web — github.com/N4khlA/Race_condition_exploiter

- Built a full-stack TOCTOU training lab with **5 real race-condition vulnerabilities** (double-spend transfers, coupon reuse, overselling, duplicate signup, reset-token reuse) behind a working storefront
- Wrote an exploit harness with **4 concurrency strategies**: asyncio, barrier-synced threads, raw TCP sockets and HTTP/2 single-connection multiplexing
- Implemented **3 detectors** (request-window grouping, PostgreSQL invariant triggers via LISTEN/NOTIFY, burst fingerprinting) and toggleable fixes (CAS, row locks, serializable, Redis locks), benchmarked for exploit rate against latency and deadlocks

`FastAPI · PostgreSQL · Redis · React + TypeScript · WebSocket · Docker Compose`

ML Network Intrusion Detection · Defensive / ML — github.com/N4khlA/AI-_threat_hunting

- Trained a grid-searched Random Forest (**macro F1 0.948**) and a bidirectional LSTM (**AUC 0.998**) on UNSW-NB15 to detect DoS, scans, exploits, fuzzing and botnet traffic
- Compared the models with a McNemar significance test, ran a leakage scan and per-attack-family recall, and documented the weak spots
- Served predictions via FastAPI (**~700k flows/s** on CPU), added a live Streamlit monitor and a retraining pipeline that refuses to replace the current model with a worse one, all covered by **49 tests**

`Python · scikit-learn · TensorFlow · FastAPI · Streamlit · SQLite · pytest`

masa: Mobile App Security Analyzer · AppSec / Mobile — github.com/N4khlA/MobileApp_SecAnalyzer

- Built a static scanner for Android APKs and iOS IPAs that scores apps against the **OWASP Mobile Top 10 (2023)**, M1–M10
- Wrote **dependency-free parsers** for binary AXML, binary plist, DEX string tables, ELF and Mach-O, including hardening checks (PIE, RELRO, executable stack)
- Detects hardcoded secrets, TrustAll managers, ATS bypasses, weak crypto and insecure storage, and generates HTML, Markdown and JSON reports

`Python standard library · OWASP MASVS · ELF / Mach-O · MIT licence`

TECHNICAL SKILLS

Offensive	Web exploitation, race conditions / TOCTOU, custom exploit tooling, HTTP/2 and raw-socket attacks
Defensive	ML-based intrusion detection, detection signals and triggers, model evaluation and drift monitoring
AppSec	OWASP Top 10 and Mobile Top 10, static analysis, binary hardening, concurrency-safe database fixes
Languages	Python, TypeScript, SQL
Tools & platforms	FastAPI, React, PostgreSQL, Redis, Docker, Linux (Kali), Git, pytest, scikit-learn, TensorFlow